Manufacturing Operational Insights Report

Real Estate Tutor Bot/ Quiz Bot

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI-27

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Academic Year: Jan-Jun 2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement and Objectives

1. Problem Statement

The real estate field includes complex topics such as property valuation, legal compliance, listings, and investment strategies. Traditional learning methods like books and static courses are not interactive and do not provide instant feedback or personalized guidance.

Learners often face difficulties in:

- Understanding practical concepts
- Identifying weak areas
- Tracking their learning progress

Hence, there is a need for an intelligent and interactive system that improves learning efficiency using modern AI technologies.

2. Project Objectives

The objective of this project is to develop an AI-driven educational quiz and tutor system for the real estate domain, specifically focused on *property listings*. The system aims to assess a user's understanding of key real estate concepts while simultaneously enhancing their knowledge through intelligent feedback.

The project is designed to move beyond traditional quiz systems by integrating a Large Language Model (LLM) to evaluate user responses and provide contextual explanations, real-world insights, and professional tips. This helps transform the application from a simple assessment tool into an interactive learning platform.

The key objectives of the project are:

- To create an interactive quiz system covering essential real estate topics such as MLS, property valuation, listing agreements, disclosures, and pricing strategies.
- To implement AI-based answer evaluation that not only checks correctness but also explains concepts in detail.
- To design a personalized learning experience by allowing users to select topics and difficulty levels.
- To identify weak areas in user performance and provide targeted re-quiz opportunities for improvement.

- To simulate real-world exam conditions using features like a question timer.
- To store user performance data and enable progress tracking over multiple attempts.
- To generate a downloadable PDF report containing detailed results and explanations for offline revision.

3. Scope of Project

- Develop an **AI-powered quiz and tutor bot** for the real estate domain.
- Cover key topics like **property listings, valuation, MLS, disclosures, and pricing strategies**.
- Provide an **interactive web-based interface** using Streamlit.
- Implement **AI-based answer evaluation** with detailed explanations and pro tips.
- Enable **personalized learning** through topic and difficulty selection.
- Include a **timer-based quiz system** to simulate real exam conditions.
- Analyze performance to identify **weak topics**.
- Provide **targeted re-quiz functionality** for weak areas.
- Store user data and quiz results using a **SQLite database**.
- Allow users to **track progress across multiple attempts**.
- Generate a **downloadable PDF report** for offline review.
- Create a system that supports both **knowledge assessment and learning improvement**.

Proposed Solution

1. Key Features

- **AI-Powered Answer Evaluation :** The system uses a Large Language Model (LLM) to evaluate the user's answer. Instead of just checking whether the answer is correct or incorrect, the AI analyzes the response and provides a more intelligent evaluation along with educational feedback.

- **Contextual Explanation Generation:** After each question, the AI generates a detailed explanation of the concept being tested. If the user answers incorrectly, the system explains why the answer is wrong and why the correct answer is correct, often using real-world real estate examples.

- **Personalized Learning Loop:** The system follows a learning cycle where users take a quiz, receive feedback, identify weak areas, and practice those areas again. This continuous process helps improve the user's understanding over time.

- **Topic & Difficulty Selection:** Before starting the quiz, users can choose specific topics and difficulty levels they want to practice. This allows users to focus on areas that are relevant to them or match their learning level.

- **Weak Topic Detection & Re-Quiz:** After the quiz is completed, the system analyzes the user's performance and identifies topics where the score is below a certain threshold. The user can then start a new quiz that focuses only on those weak topics.

- **Question Timer:** Each quiz question has a time limit (e.g., 30 seconds). This feature simulates real certification exams and encourages users to answer questions within a limited time.

- **Performance Dashboard:** At the end of the quiz, the system displays a summary dashboard that shows the user's results, including accuracy, grade, topic-wise performance, and difficulty-level performance.

- **PDF Report Export:** The application can generate a downloadable PDF report containing the quiz results, explanations, and performance analysis. This allows users to review their mistakes and study the concepts later.

- **Multi-User Login System:** The system supports multiple users, allowing different people to log in using their names. Each user has their own quiz history and performance records.

- **Persistent Score History (SQLite):** The project uses a SQLite database to store user information and quiz attempts. This allows the system to save results permanently and track user progress across multiple quiz sessions.

2. Overall Architecture / Workflow

The RealEstate IQ Bot follows a modular client-server architecture where all components run locally as a Python-based web application. The system is divided into five layers — Presentation, Application Logic, AI Evaluation, Data Persistence, and External Services — each with a distinct responsibility.

Architecture Layers:-

- **Presentation Layer** — Built using Streamlit, the UI consists of six screens: Login, Welcome, Filter, Quiz, Summary, and History. All screen navigation is controlled through a central phase-based router in app.py using Streamlit's session_state.

- **Application Logic Layer** — Business logic is distributed across four modules. session_manager.py manages quiz state, score tracking, and weak topic detection. question_bank.py holds all 20 domain-specific questions with metadata. evaluator.py constructs prompts and handles API communication. pdf_generator.py builds downloadable reports in memory.

- **AI Evaluation Layer** — When a user submits an answer, evaluator.py sends a structured prompt to the Groq API containing the question, all four options, the correct answer, the student's answer, topic, difficulty, and an educator hint. The Llama 3.3 70B model returns a JSON object containing a correctness flag, score, key concept, contextual explanation, and professional tip. Retry logic handles rate limit errors automatically.

- **Data Persistence Layer** — database.py manages all interactions with a local SQLite database containing three tables — users, quiz_attempts, and question_responses. Data is written once per completed quiz and read on demand for the history screen and summary dashboard.

- **External Services** — The only external dependency is the Groq API. All other functionality is self-contained within the local application.

Overall Workflow:-

```
User opens app
       ↓
Login / Register → credentials verified against SQLite users table
       ↓
Welcome screen → user sees past attempt count and navigation options
       ↓
Filter screen → user selects topics and difficulty levels
       ↓
Quiz screen → one question at a time with 30-second countdown timer
       ↓
Each answer → sent to Groq API → AI returns score + explanation
       ↓
Result panel → correct/incorrect feedback + AI explanation + pro tip
       ↓
After all questions → attempt saved to SQLite automatically
       ↓
Summary dashboard → accuracy, grade, topic breakdown, weak areas
       ↓
Weak topics found → "Practice Weak Topics" button starts re-quiz
No weak topics → "Retake Quiz" or "View History"
       ↓
PDF export → full report generated in memory and downloaded
```

Key Design Decisions:-

- **Separation of concerns** — Each module has exactly one responsibility, making it straightforward for different team members to work on different parts simultaneously without conflicts.

- **Stateless AI calls** — The Groq API is called once per question with complete context in every request. The model has no memory between calls — all state is managed by the application, not the AI.

- **In-memory PDF generation —** Reports are built using io.BytesIO() and never written to disk, keeping the application clean and portable.

- **Phase-based routing —** All navigation is driven by a single phase variable in session_state, making the control flow easy to trace and debug.

3. Tools and Technologies used

Programming Language

- Python 3.10+

Frontend / UI

- Streamlit — web-based interactive UI framework

AI / LLM

- Groq API — cloud inference platform
- Llama 3.3 70B — open-source LLM used for answer evaluation and explanation generation

Database

- SQLite — lightweight local relational database (built into Python)

PDF Generation

- ReportLab — Python library for programmatic PDF creation

Security

- python-dotenv — secure API key management via .env file
- hashlib — built-in Python library for SHA-256 password hashing

IDE & Development

- PyCharm — primary development environment
- Git — version control
- GitHub — remote repository hosting

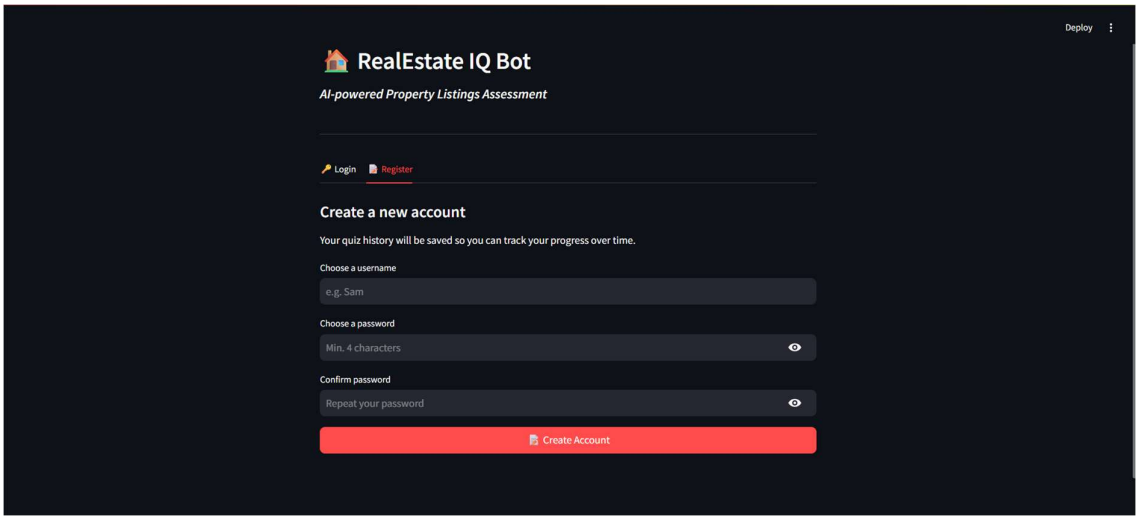
Key Python Libraries

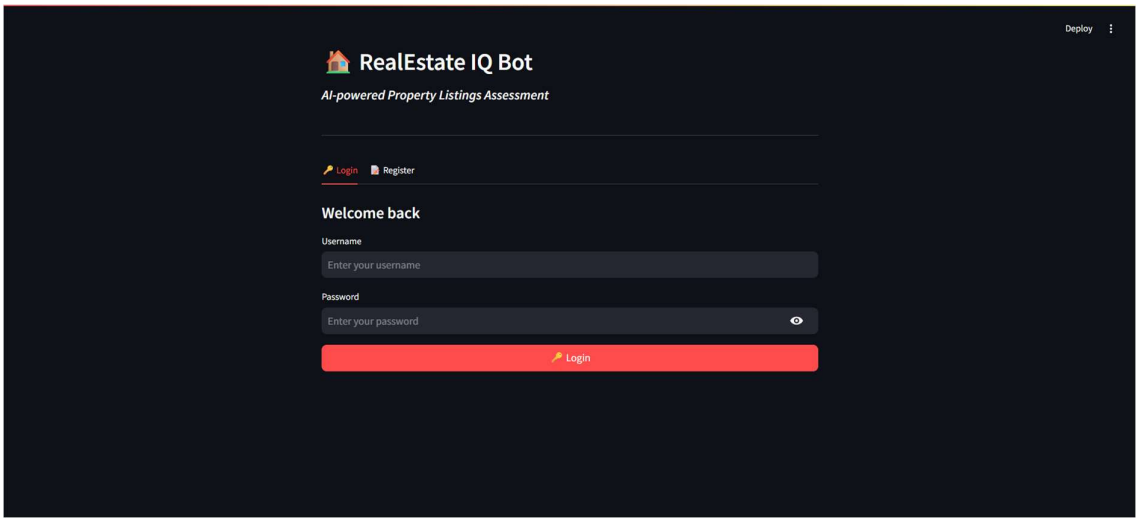
- sqlite3 — database access (built-in)
- groq — official Groq Python SDK
- io — in-memory PDF byte stream (built-in)
- json — API response parsing (built-in)
- hashlib — password hashing (built-in)
- time — question timer logic (built-in)

Results and Output

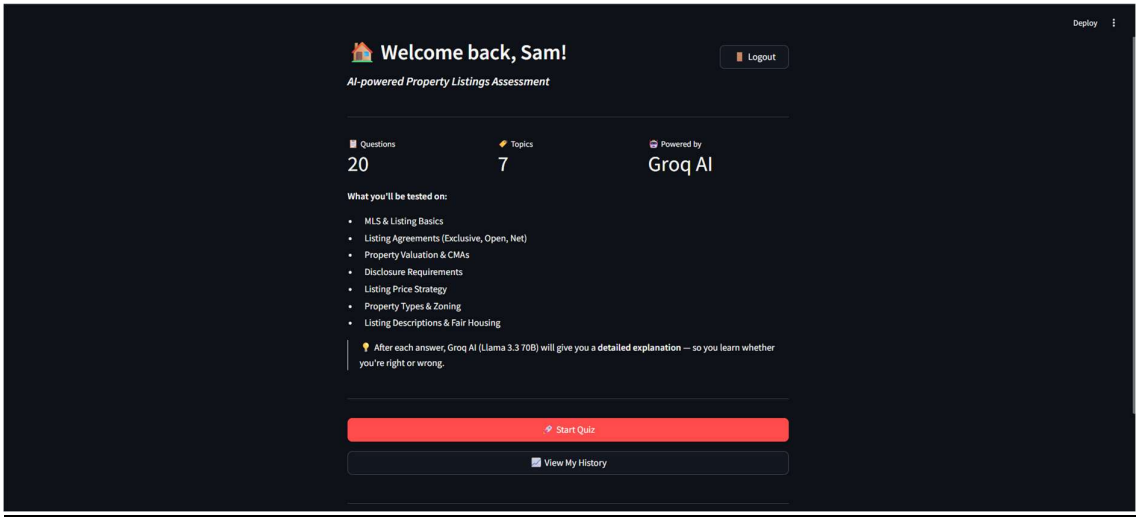
1. Screenshots / Outputs

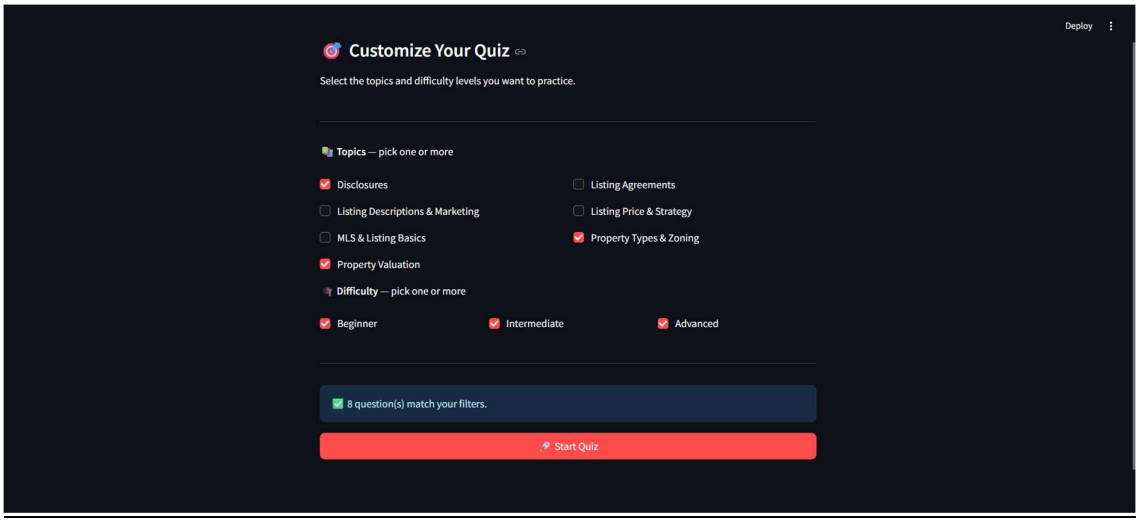
Login / Register Screen:-



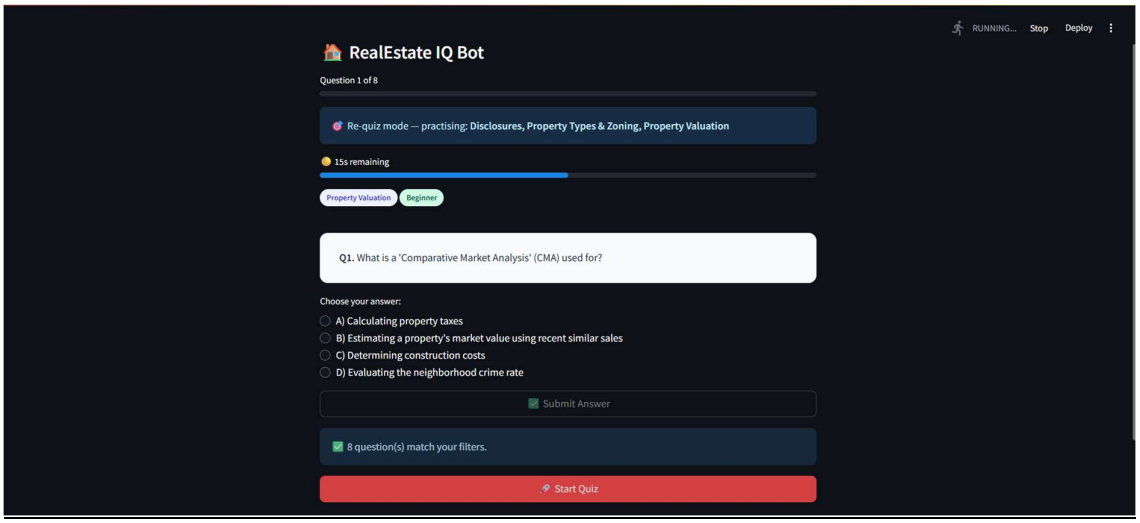


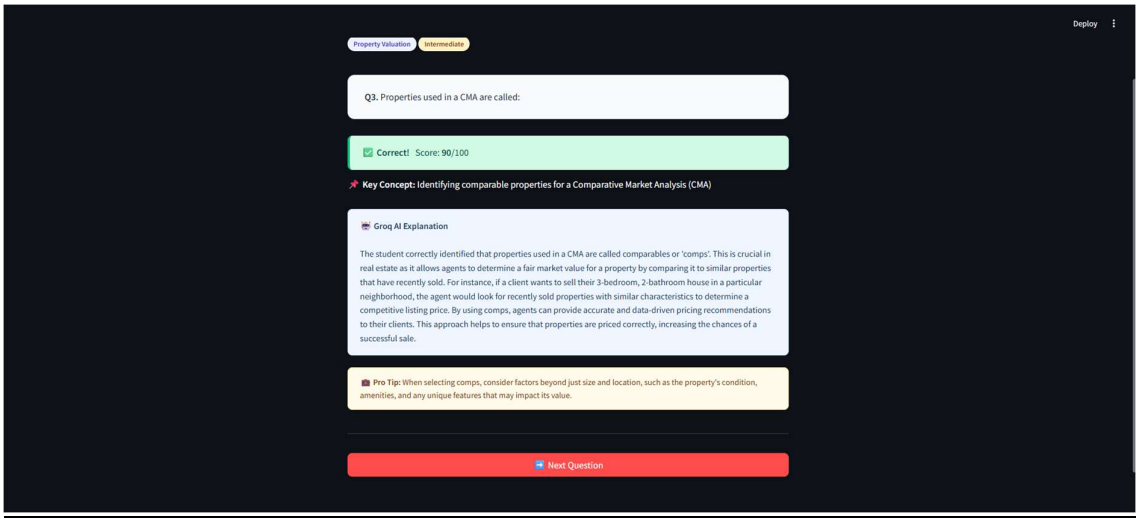
Welcome screen:-



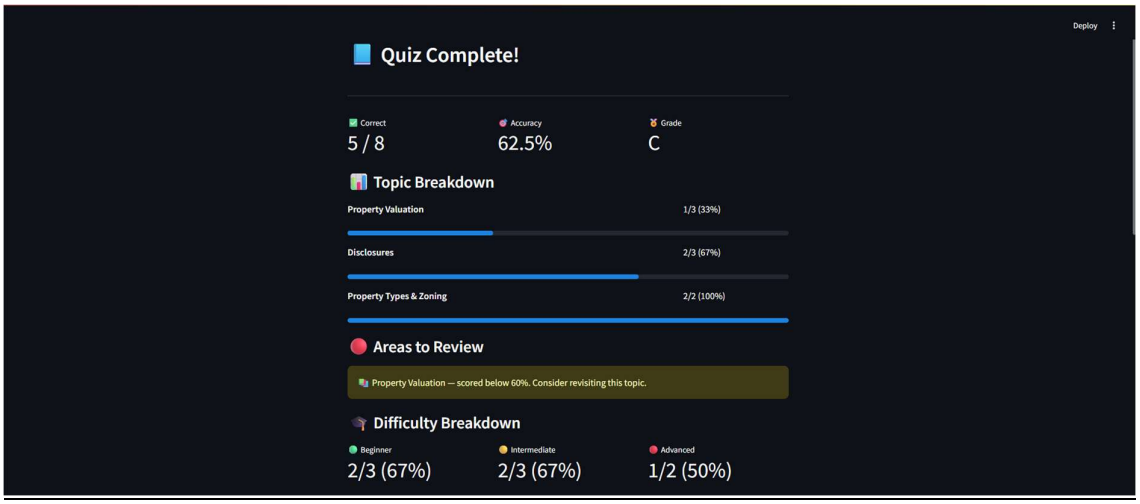


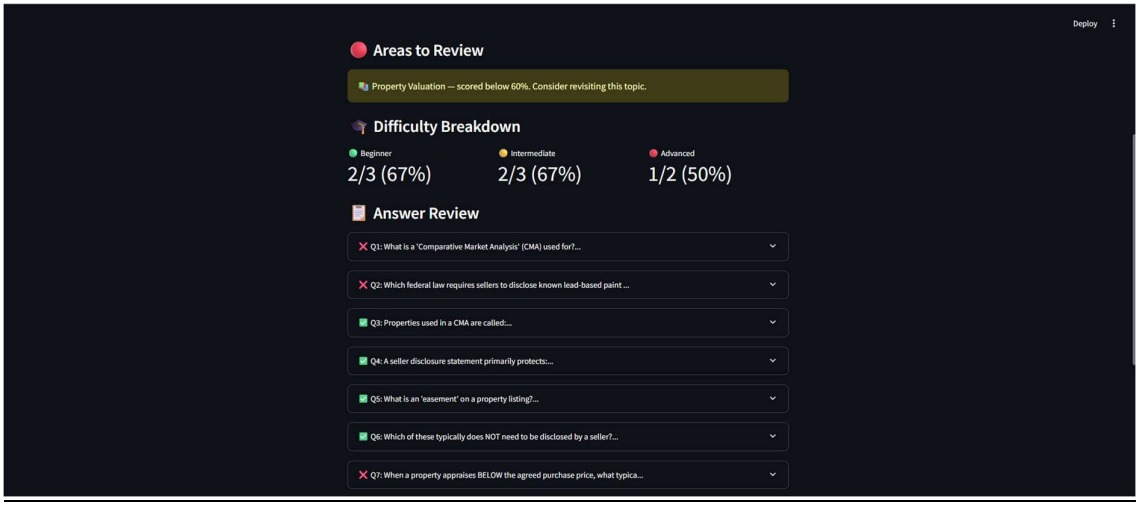
Quiz Screen:-

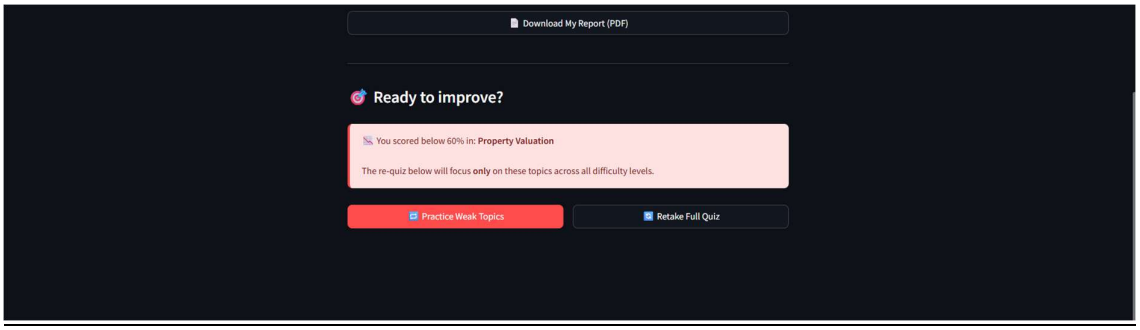

RUNNING... Stop Deploy
RealEstate IQ Bot
Question 1 of 8
Re-quiz mode — practising: Disclosures, Property Types & Zoning, Property Valuation
15s remaining
Property Valuation
Beginner
Q1. What is a 'Comparative Market Analysis' (CMA) used for?
Choose your answer:
A) Calculating property taxes
B) Estimating a property's market value using recent similar sales
C) Determining construction costs
D) Evaluating the neighborhood crime rate
Submit Answer
8 question(s) match your filters.
Start Quiz


Deploy
Property Valuation
Intermediate
Q3. Properties used in a CMA are called:
Correct! Score: 90/100
Key Concept: Identifying comparable properties for a Comparative Market Analysis (CMA)
Groq AI Explanation
The student correctly identified that properties used in a CMA are called comparables or 'comps'. This is crucial in real estate as it allows agents to determine a fair market value for a property by comparing it to similar properties that have recently sold. For instance, if a client wants to sell their 3-bedroom, 2-bathroom house in a particular neighborhood, the agent would look for recently sold properties with similar characteristics to determine a competitive listing price. By using comps, agents can provide accurate and data-driven pricing recommendations to their clients. This approach helps to ensure that properties are priced correctly, increasing the chances of a successful sale.
Pro Tip: When selecting comps, consider factors beyond just size and location, such as the property's condition, amenities, and any unique features that may impact its value.
Next Question

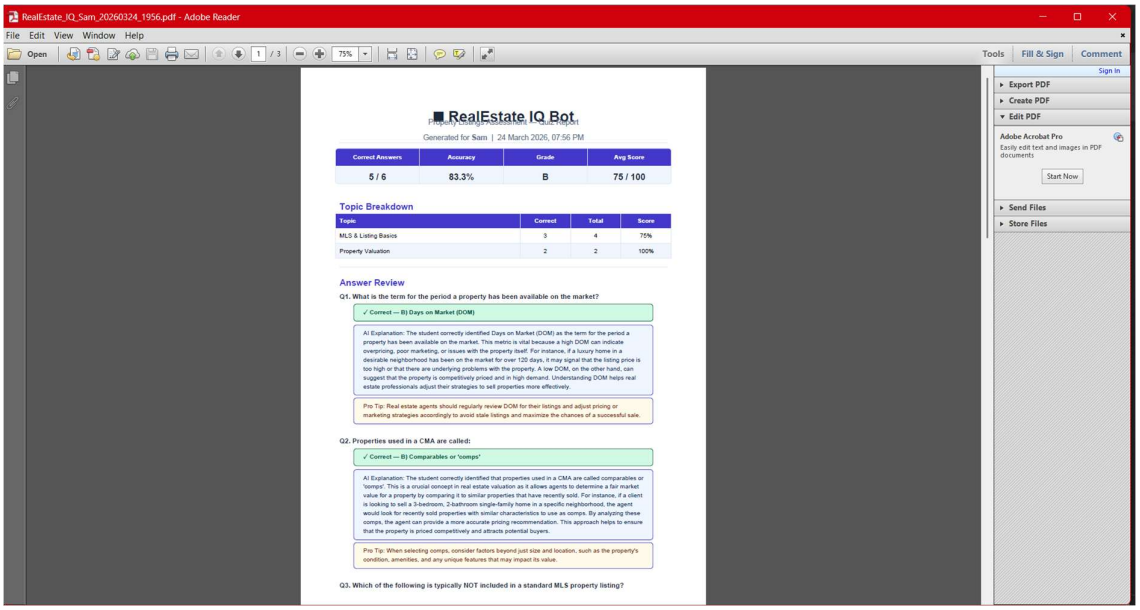
Result / Summary Screen:-







Downloaded pdf preview:-



2. Key Outcomes

- Successfully developed a fully functional AI-powered educational bot for the Real Estate domain
- Integrated Groq's Llama 3.3 70B model to provide contextual, domain-specific explanations for every answer
- Implemented a personalized learning loop that detects weak topics and automatically routes users to re-practice them
- Built a multi-user system with secure login, password hashing, and isolated score histories
- Achieved persistent score tracking across sessions using SQLite, enabling long-term progress monitoring
- Generated downloadable PDF reports containing AI explanations, useful for offline revision
- Delivered a complete 6-screen web application with zero frontend coding using Streamlit
- Successfully migrated across three AI providers (Claude → Gemini → Groq) with minimal code changes, demonstrating clean modular architecture
- Covered 7 real estate topics and 3 difficulty levels across 20 curated domain-specific questions
- Implemented exam-like conditions with a 30-second per-question countdown timer and auto-submission on expiry

Conclusion

The RealEstate IQ Bot was successfully designed and developed as an AI-driven educational assessment platform focused on the Real Estate domain, specifically Property Listings. The project fulfills its core problem statement by moving beyond traditional quiz applications — instead of simply marking answers right or wrong, the system leverages a Large Language Model to generate contextual, domain-specific explanations that actively teach the user through every interaction.

The application demonstrates a complete GenAI product lifecycle — from prompt engineering and structured LLM output parsing, to session management, persistent data storage, multi-user authentication, and professional report generation. The modular architecture ensured clean separation of concerns, allowing different team members to work independently on different components without conflicts.

Key engineering challenges encountered and resolved during development included AI provider migration, rate limit handling, Streamlit session state management, and database schema design — all of which contributed to practical, industry-relevant learning outcomes for the team.

The project successfully bridges the gap between theoretical real estate knowledge and active learning, making it a practical tool for students, trainees, and professionals preparing for real estate certification examinations.

Future Scope and Enhancements

Content Expansion

- Expand the question bank beyond 20 questions to cover additional real estate domains such as Commercial Properties, Real Estate Law, Mortgage & Finance, and Property Management
- Add image-based questions using actual property listing photos

AI Enhancements

- Integrate RAG (Retrieval Augmented Generation) to allow users to upload real estate handbooks or MLS documentation and generate questions dynamically from the content
- Add a conversational Study Mode where users can ask free-form questions between quiz attempts
- Implement adaptive difficulty — the AI automatically adjusts question difficulty based on the user's historical performance

Platform Enhancements

- Deploy on Streamlit Cloud to make the application publicly accessible via a URL without any local installation
- Add a leaderboard to introduce competitive learning among multiple users
- Implement email notifications that send weekly performance summaries to registered users

Data & Analytics

- Build an admin dashboard showing aggregate performance metrics across all users and topics
- Add visual progress charts showing accuracy trends over time using Plotly or Matplotlib
- Implement spaced repetition — questions the user got wrong reappear after a calculated interval for optimum retention

Assessment Features

- Add a certification mode with a formal pass/fail threshold and a printable certificate on passing
- Support timed full-length mock exams simulating real real estate licensing test conditions